

NUS-ISS Intelligent Reasoning Systems

Practice Module

FleetCare AI

A Hybrid Temporal Reasoning System for Predictive Maintenance of Heavy-Duty Trucks

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1 Introduction

FleetCare AI proposes an intelligent predictive maintenance decision-support system for heavy-duty vehicles. Using historical operational readouts and vehicle specifications from the SCANIA Component X dataset, the project combines machine learning with temporal risk analysis and rule-based reasoning to identify vehicles that may require maintenance attention and provide interpretable, evidence-based recommendations. The system is designed to help fleet and maintenance managers answer three connected questions: which vehicles require attention, what evidence supports their prioritisation, and what level of maintenance action should be considered.

The central challenge is to connect predictive information with maintenance decisions. A predicted repair-proximity state can indicate the urgency of a vehicle's condition, but a single prediction does not show whether the signal has persisted, whether the vehicle is deteriorating, or whether sufficient historical evidence supports escalation. Relying on prediction alone may therefore provide an incomplete basis for maintenance prioritisation.

FleetCare AI addresses this challenge by treating predictive outputs as evidence for a broader reasoning process rather than as maintenance decisions themselves. Current predictions are considered together with historical patterns and temporal changes to assess the strength and development of maintenance risk. Explicit reasoning is then used to translate this evidence into prioritised and interpretable maintenance recommendations. This separation between prediction and decision support is intended to make the basis of each recommendation more transparent and easier for maintenance professionals to inspect.

The project has four objectives:

- Develop a vehicle-aware predictive approach that combines current and historical operational information to estimate repair-proximity states and multi-horizon risk scores.
- Incorporate temporal evidence to distinguish persistent or worsening risk from isolated predictive signals.
- Design transparent maintenance reasoning that translates predictive and historical evidence into prioritised recommendations with explicit supporting evidence.
- Provide an interactive interface through which fleet and maintenance managers can inspect vehicle assessments, recommendations, explanations, and relevant supporting knowledge.

FleetCare AI is intended to support rather than automate maintenance decision-making. Inspection and repair decisions remain the responsibility of fleet and maintenance professionals.

2 Project Background

2.1 Problem Background

Heavy-duty vehicle fleets operate under demanding and variable conditions, making maintenance an important factor in vehicle availability and fleet operations. Unexpected repair needs can result in unplanned downtime and disrupt vehicle utilisation, while unnecessary or poorly prioritised maintenance can consume limited maintenance capacity. Fleet and maintenance managers must therefore decide not only whether a vehicle may require attention, but also which vehicles should be prioritised when multiple vehicles exhibit signs of potential maintenance need.

Conventional maintenance strategies are commonly based on reactive maintenance after a problem becomes apparent or preventive maintenance performed according to predefined schedules. While these approaches remain important, they may not fully reflect differences in how individual vehicles are operated or how their condition develops over time. The increasing availability of operational vehicle data creates an opportunity to complement these approaches with predictive maintenance, where historical patterns are used to identify emerging maintenance needs before a repair event occurs.

This problem is particularly relevant at fleet level. A maintenance manager may need to assess many vehicles simultaneously, each with different operating histories and levels of available evidence. Identifying a potentially high-risk vehicle is therefore only part of the problem; maintenance planning also requires vehicles to be prioritised according to the strength and urgency of the available evidence.

The SCANIA Component X dataset used in this project provides a basis for investigating this problem using longitudinal heavy-duty vehicle data. It contains operational readouts and vehicle specification attributes for a fleet of vehicles, with multiple observations available for individual vehicles at irregular operational intervals. The prediction task is to estimate a vehicle's repair-proximity state at its latest available observation. Five classes represent progressively shorter intervals to an observed repair event:

Table 1: Five repair-proximity classes used in FleetCare AI and their interpretations. Intervals are expressed in operational time units rather than calendar time.

Class	Repair-proximity interval	Interpretation in FleetCare AI
C0	No repair within 48 time units	No near-term repair observed
C1	24–48 time units	Early repair proximity
C2	12–24 time units	Elevated repair proximity
C3	6–12 time units	Near-term repair proximity
C4	0–6 time units	Most imminent repair state

This formulation introduces several modelling challenges. First, the class distribution is highly imbalanced, with most observations belonging to C0 and relatively few examples in C1–C4. Second, vehicle histories vary in length and are recorded at irregular operational intervals. Third, some histories are censored because no repair is observed before the end of the study period. These characteristics complicate both model development and reliable evaluation.

The longitudinal structure introduces an additional challenge: vehicles with similar current observations may have substantially different preceding trajectories, while the amount and recency of available history also vary across vehicles. Historical context may therefore contain information that is not represented by the latest observation alone.

2.2 Predictive Maintenance and Decision-Support Context

Predictive maintenance uses operational and historical data to assess equipment condition and identify potential maintenance needs before repair events occur. Machine-learning approaches are particularly useful when operational data are high-dimensional and contain complex relationships that may be difficult to represent through manually defined rules alone.

Predictive assessment and maintenance decision-making nevertheless serve different purposes. Predictive models estimate equipment condition or repair proximity, whereas decision-support systems place these estimates within the operational context required for maintenance prioritisation.

3 Market Research

The commercial fleet-management market demonstrates a clear demand for technologies that reduce unplanned downtime, improve vehicle availability, and support more proactive maintenance planning. Connected vehicles increasingly generate operational and diagnostic data that can support data-driven fleet operations and predictive maintenance [1].

Major commercial platforms demonstrate this shift toward connected and predictive maintenance. Scania Digital Services uses connected-vehicle data to support vehicle monitoring, service planning, and proactive maintenance [7]. Geotab provides fleet analytics, vehicle diagnostics, predictive maintenance insights, and maintenance workflows [3]. Samsara integrates real-time diagnostics and fault monitoring with maintenance scheduling and operational workflows [6]. Beyond vehicle fleets, IBM Maximo demonstrates the broader industrial adoption of predictive asset-health and maintenance management [4].

Together, these solutions demonstrate that connected monitoring, predictive analytics, and maintenance workflow support are established directions in fleet and asset management. They also confirm the practical relevance of using operational vehicle data to support more proactive maintenance decisions.

FleetCare AI therefore does not seek to introduce predictive maintenance as a new concept. Instead, the project investigates a more specific decision-support problem: how predictive evidence can be translated into a transparent and inspectable maintenance recommendation. The project focuses on the connection between predictive assessment and maintenance prioritisation, where current model outputs may be considered together with temporal and fleet-relative evidence before a maintenance action is recommended.

Academic research provides further support for this direction. Machine-learning methods have been widely investigated for predictive maintenance [2], including automotive applications in which vehicle operating data are used to support data-driven maintenance assessment [8]. Explainability techniques such as SHAP provide a mechanism for examining the contribution of individual features to model predictions [5]. These established approaches provide a foundation for predictive assessment and model interpretability.

Table 2: Selected commercial fleet and asset-management solutions and their relevance to FleetCare AI.

Existing Solution	Key Strength	Relevance to FleetCare AI
Scania Digital Services	Connected-vehicle monitoring and proactive service planning	Demonstrates the operational value of longitudinal vehicle data for proactive maintenance planning
Geotab	Fleet analytics, diagnostics, and predictive maintenance	Demonstrates demand for data-driven identification and prioritisation of fleet maintenance needs
Samsara	Real-time diagnostics integrated with maintenance workflows	Demonstrates the value of connecting vehicle-condition information with operational maintenance processes
IBM Maximo	Predictive asset-health and maintenance management	Demonstrates the wider industrial adoption of predictive information for maintenance decision support

3.1 Positioning of FleetCare AI

Against this commercial and technical context, FleetCare AI is positioned as an intelligent predictive maintenance decision-support prototype rather than an autonomous maintenance or failure-diagnosis system. It is intended to complement existing predictive-maintenance and fleet-management approaches by focusing on the connection between predictive assessment and maintenance prioritisation.

The project's differentiation lies in investigating an interpretable bridge between predictive modelling and maintenance decision support. Predictive assessment provides evidence of repair proximity, temporal analysis provides context on how that evidence develops over time, and explicit reasoning provides a transparent mechanism for translating the combined evidence into a maintenance recommendation. This enables the predictive output and the reasoning behind the recommended action to be inspected separately.

The resulting recommendations are intended to support professional maintenance judgement rather than replace it.

4 Project Scope

The project scope covers four system capabilities and their integration within a predictive-maintenance decision-support prototype.

4.1 System Capabilities

The project covers four main capabilities:

1. Repair-Proximity Prediction
   A vehicle-aware multi-history machine-learning model estimates the five repair-proximity states (C0–C4) defined in Table 1.
2. Historical Risk Assessment
   Model class scores are aggregated into four cumulative repair-horizon risk scores: Risk48, Risk24, Risk12, and Risk6. Historical model outputs are used to assess whether elevated risk is persistent, whether predicted repair proximity is worsening, and whether risk is deteriorating unusually quickly relative to other vehicles in the fleet. This provides historical context for the current prediction rather than relying on a single model output.
3. Knowledge-Based Maintenance Reasoning
   A transparent rule-based reasoning layer combines the current predicted state with fleet-relative risk and historical evidence, including risk persistence, relative deterioration, recent class trajectory, and history sufficiency. It converts this evidence into maintenance recommendations such as Continue, Monitor, Enhanced Monitor, Inspect, Priority Inspect, or Manual Review.

4. Explainable Decision Support
   SHAP will be used to identify the features contributing to the machine-learning prediction, while explicit reasoning traces will explain why a particular maintenance recommendation is generated. The resulting information will be structured for presentation through a fleet-management dashboard and interaction through a RAG-based assistant.

4.2 Intelligent Reasoning Techniques

The project adopts a hybrid intelligent reasoning approach combining:

Machine Learning + Historical Risk Evidence + Knowledge-Based Reasoning + Explainability

These techniques perform complementary reasoning functions. Machine learning provides a data-driven assessment of repair proximity, historical analysis evaluates how predictive evidence develops over time, and knowledge-based rules translate the combined evidence into an explicit maintenance recommendation. Explainability is applied to both the predictive and decision layers so that the basis of each assessment can be inspected.

The architecture therefore separates prediction from decision-making: the predictive component estimates vehicle condition, while the reasoning component determines how the available evidence should influence maintenance priority.

4.3 Scope Boundaries and Limitations

The project is designed as a decision-support prototype rather than an autonomous maintenance system. Several limitations define its current scope:

- Anonymised features: Operational variables and vehicle specifications are anonymised, limiting the physical interpretation of individual model features and SHAP explanations.
- Severe class imbalance: C0 vehicles dominate the dataset, while C1–C4 examples are rare, making minority-class prediction challenging.
- Censored observations: For vehicles without an observed repair, the actual repair time beyond the observation period is unknown. Censoring must therefore be explicitly considered during training-label construction.
- Model risk scores: Risk48, Risk24, Risk12, and Risk6 are treated as model-derived risk scores rather than calibrated real-world probabilities of repair. Probability calibration will be considered during model evaluation.
- Prototype maintenance rules: Rule thresholds are designed as transparent prototype decision policies and have not been validated as official Scania maintenance procedures or by domain experts.
- Decision-support use: Recommendations are intended to prioritise vehicles for attention and support maintenance professionals rather than automatically trigger inspection or repair.
- Limited component-level diagnosis: Because the operational features are anonymised, the system can identify influential operational patterns but cannot determine which physical component should be inspected or repaired.
- Market applicability: Deployment in a real fleet would require validation using live operational data, integration with existing maintenance systems, domain-expert validation of the decision rules, and continued monitoring for data and model drift.

5 Data Collection and Preparation

FleetCare AI uses historical vehicle data to learn repair-proximity patterns and documented knowledge to support interpretable maintenance reasoning. Data preparation connects these sources to the proposed system by constructing suitable training labels, representing vehicle history, and separating predictive inputs from information available only for training and evaluation.

Table 3: Dataset inventory recomputed from the supplied operational CSV files. Readouts are repeated observations, whereas vehicles are distinct entities within each partition.

Partition	Vehicles	Operational readouts
Training	23,550	1,122,452
Validation	5,046	196,227
Test	5,045	198,140

5.1 Data Source and Characteristics

The project uses the publicly available SCANIA Component X dataset, collected by Scania CV AB and distributed through Researchdata.se under the Creative Commons Attribution 4.0 International licence.[1] The dataset combines longitudinal operational measurements, vehicle specifications, and repair-related information for an anonymised truck engine component. The accompanying data descriptor is provided by Kharazian et al. (2025).[2]

Data organisation and analysis scope. Exploratory analysis was conducted directly on the supplied CSV files, following the data-inspection dimensions identified in the project's exploration script: vehicle coverage, readout frequency, repair status, and missingness. The statistics below describe the source data before imputation, sampling, or feature expansion; they are not model evaluation results. Detailed temporal and missingness analysis uses the training partition, while the supplied validation labels are summarised separately to describe the classification task. Test labels are not analysed here.

The operational table contains 105 feature columns, excluding vehicle identifiers and time. Their prefixes identify 14 anonymised variable groups; multiple columns can therefore belong to the same underlying variable rather than represent independent sensors. Eight additional vehicle-specification fields provide static context. Training vehicle identifiers match across the operational, specification, and repair tables. No duplicate vehicle–time pairs were found in the training readouts, and neither the validation nor test population overlaps with the training vehicle identifiers. These checks support reliable table joins and preservation of the supplied partition boundaries.

Repair-event imbalance and censoring. Of the 23,550 training vehicles, 2,272 (9.65%) have an observed in-study repair and 21,278 (90.35%) do not. Figure 1(a) shows this vehicle-level imbalance. Absence of an observed repair does not establish that a vehicle remains repair-free after observation ends. Such records are right-censored: their subsequent repair time is unknown. Consequently, a readout should only receive a no-repair-within-horizon label when sufficient follow-up is available; otherwise, its status for that horizon remains uncertain.

The validation labels show a more pronounced imbalance across repair-proximity states: C0 contains 4,910 vehicles (97.30%), while C1–C4 contain 16, 14, 30, and 76 vehicles, respectively. These labels describe proximity at the validation assessment point, unlike the training repair indicator, which records whether a repair occurred during the study. They should therefore not be interpreted as evidence of a change in repair incidence between partitions. The small minority-class counts motivate class-sensitive evaluation and cautious interpretation of subsequent estimates, rather than reliance on overall accuracy alone.

Longitudinal coverage and irregular sampling. Training vehicles contribute between 5 and 303 readouts, with a median of 43 and an interquartile range of 28–64 (Fig. 2). Treating every readout as an equally independent sample would give greater influence to vehicles with longer or denser histories. This observation motivates vehicle-aware preparation and explicit separation of vehicle-level and readout-level statistics.

Consecutive-readout intervals have a median of 4.4 operational time units and an interquartile range of 1.6–6.2; the maximum is 387.8. Thus, a fixed number of previous readouts does not represent a fixed duration. Historical features should account for elapsed operational time, and temporal reasoning should recognise gaps in the available evidence. The dataset's time scale is retained throughout the project without converting its units into calendar days or assigning an unsupported physical interpretation.

[1] Official dataset record: https://researchdata.se/en/catalogue/dataset/2024-34.

[2] Kharazian et al., "SCANIA Component X dataset: a real-world multivariate time series dataset for predictive maintenance," *Scientific Data*, 12, 493 (2025). https://doi.org/10.1038/s41597-025-04802-6.

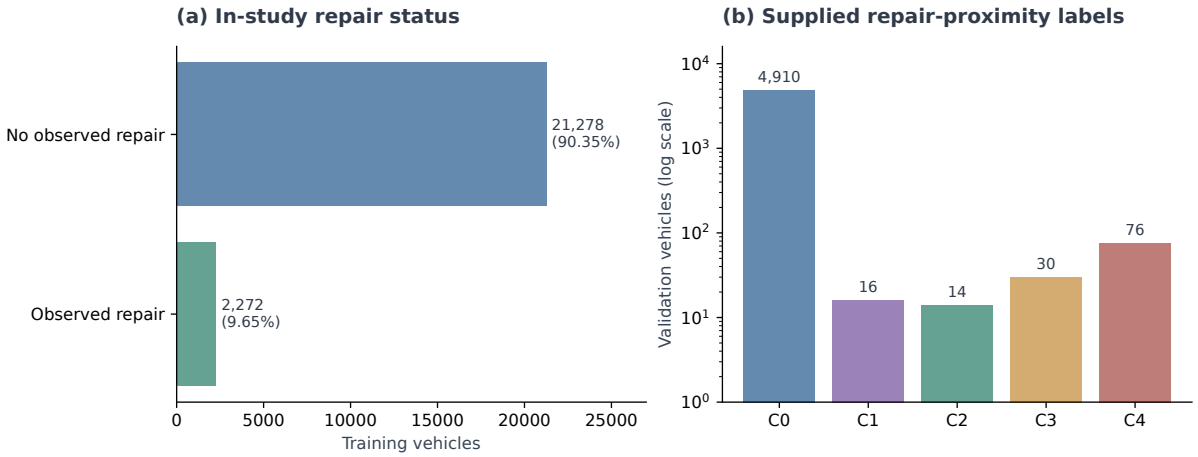


Fig. 1: Two distinct measures of data imbalance: (a) observed repair status among training vehicles and (b) supplied C0–C4 labels among validation vehicles. Panel (b) uses a logarithmic count axis to make minority classes visible. These are source-data counts, not predicted classes; the two panels describe different populations and label definitions.

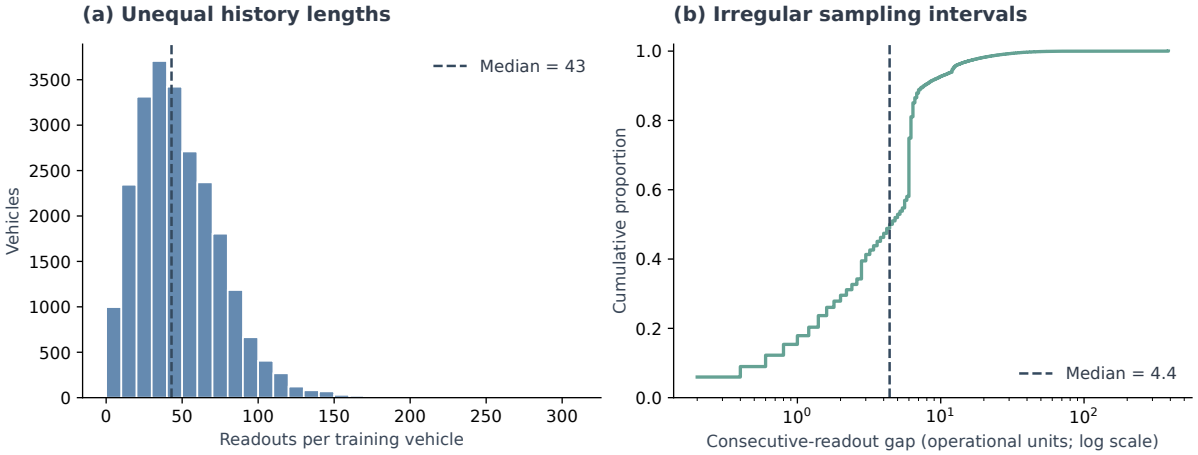


Fig. 2: Temporal coverage in the training data: (a) distribution of readout counts across 23,550 vehicles and (b) cumulative distribution of 1,098,902 within-vehicle consecutive-readout intervals after sorting by time. Panel (b) uses a logarithmic horizontal axis and includes the full interval range. Dashed lines indicate medians.

Missingness and anonymised feature representation. Across all training operational feature cells, 0.301% are missing. At least one missing entry occurs in 104 of the 105 columns, although the largest column-level missingness is only approximately 0.858%. No missing entries were found in the eight training specification fields. Figure 3 shows both the highest-missingness columns and the distribution across all operational columns. The repeated rates among columns with the same prefix warrant considering grouped feature availability rather than assuming each column has an independent missingness pattern.

The modest overall missingness supports explicit imputation and data-quality checks without characterising the dataset as extensively incomplete. Anonymisation introduces a separate limitation: feature identifiers cannot reliably be translated into named physical measurements or failure mechanisms. Subsequent explanations will therefore describe model associations with anonymised features rather than claim a causal diagnosis.

Together, these observations establish the data-related requirements for FleetCare AI: censoring-aware targets, vehicle-aware preparation, time-aware historical features, consistent missing-value handling, and evaluation that remains sensitive to rare repair-proximity classes. They provide an empirical basis for the proposed design while leaving model effectiveness to subsequent validation.

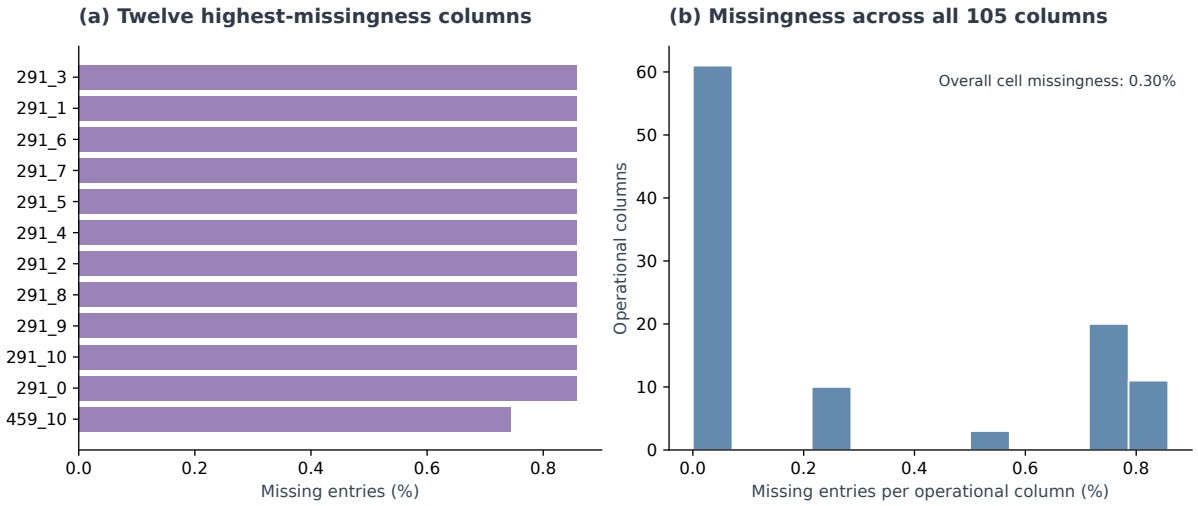


Fig. 3: Missingness before preprocessing in the training operational table: (a) twelve columns with the highest missing-entry proportions and (b) the distribution of these proportions across all 105 feature columns. Overall cell missingness is approximately 0.301%. Column names retain their original anonymised identifiers.

5.2 Data Organisation

The SCANIA Component X data are organised longitudinally by vehicle, with observations ordered according to `time_step`. Each modelling example represents a vehicle at a particular observation time and uses only information available at or before that time. Vehicle identifiers are retained for grouping, history construction, and traceability, but are excluded from predictive features.

5.3 Target Construction and Censored Observations

The supervised target represents repair proximity at a particular vehicle observation. For vehicle i with an observed repair at operational time T_i, time to repair at observation time t is

$$\mathrm{TTE}_{i,t} = T_i - t. \tag{1}$$

Eligible observations are assigned to the five repair-proximity classes defined in Table 1. Observations after a recorded repair are excluded.

For vehicles without an observed repair, future repair time beyond the available follow-up period is unknown. A censored observation is therefore assigned to C0 only when more than 48 operational time units of repair-free follow-up are available. Observations with insufficient coverage are excluded from five-class supervised training rather than assumed to be negative.

Repair records and derived time-to-repair information are used only for target construction and evaluation and are excluded from predictive features.

5.4 Leakage Control and Validation Design

The validation design is vehicle-aware and time-consistent. Training and held-out partitions are separated by vehicle, and historical features at observation time t are constructed only from information available at or before t. Future readouts, repair outcomes, and derived target information are excluded from predictive inputs.

Final evaluation produces one assessment per vehicle at its latest available readout, reflecting the intended use case of assessing a vehicle's current condition from its available history. This design reduces both vehicle-level and temporal leakage.

5.5 Data Preparation and Temporal Feature Engineering

Operational readouts are grouped by vehicle and ordered by `time_step`, while static vehicle specifications are associated with the corresponding histories. The irregular longitudinal history is transformed into a

fixed-length representation combining current operational values with previous-state features, changes between observations, time-normalised rates of change, multi-readout changes, and rolling summaries.

Time-normalised changes account for differences in elapsed operational time between readouts. Features requiring unavailable historical observations remain undefined until sufficient history is available, while history availability is retained as contextual information for downstream reasoning.

5.6 Class Imbalance and Vehicle-Aware Training

Given the severe class imbalance and variation in vehicle history length identified in Section 5.1, the proposed training strategy uses vehicle-aware sampling to control the contribution of individual vehicles while preserving examples across repair-proximity horizons. Censored vehicles contribute only observations with sufficient follow-up to support a C0 label.

Evaluation will not rely on overall accuracy alone, since performance on the dominant C0 class could mask poor identification of near-repair vehicles. Model assessment will therefore consider macro-averaged classification metrics, minority-class performance, and horizon-specific risk-ranking metrics.

6 System Design

6.1 Overall Architecture

Figure 4 presents the proposed system architecture. The workflow connects data preparation, predictive modelling, temporal evidence generation, rule-based maintenance reasoning, explanation, and user interaction. Each component has a distinct responsibility and passes structured outputs to subsequent components. This modular organisation allows predictive methods, reasoning policies, and user-facing components to be refined and evaluated independently.

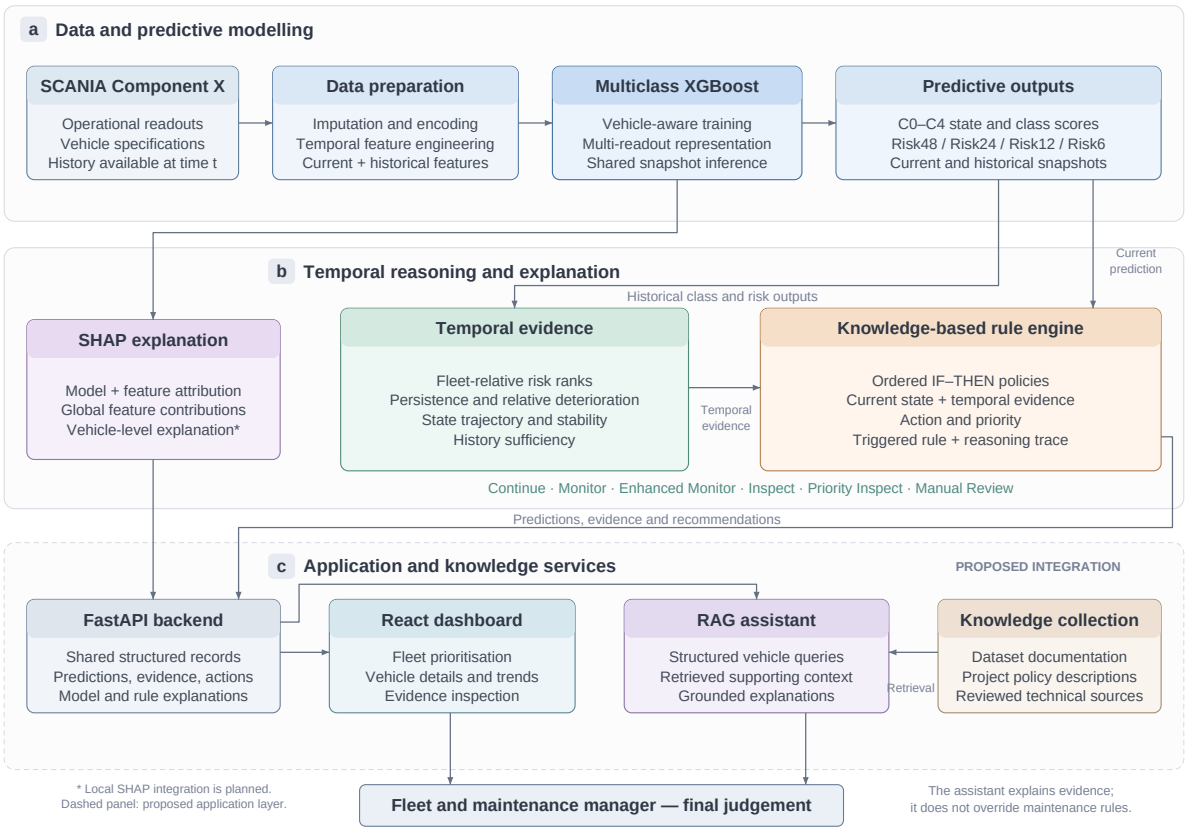


Fig. 4: Overview of the proposed FleetCare AI system.

Data preparation and prediction. Operational readouts are associated with vehicle specifications and processed chronologically. For an assessment at time t, feature construction uses only information available at or before t. The resulting vehicle representation is used to estimate one of five repair-proximity states (C0–C4) together with the corresponding class scores, from which horizon-specific risk scores are derived. These outputs describe the predictive assessment and do not directly prescribe a maintenance action.

Temporal evidence and maintenance reasoning. The current predictive output is interpreted in the context of the vehicle's recent history. Temporal analysis derives indicators such as risk persistence, deterioration, changes in predicted state, and history availability. Fleet-relative percentiles provide additional context by indicating how a vehicle's current risk and deterioration compare with the wider fleet. The rule-based reasoning component combines this evidence with the current prediction to assign a maintenance priority and recommended action, while retaining the conditions responsible for the recommendation as an explicit reasoning trace. The reasoning layer does not alter the predicted repair-proximity state, allowing maintenance policies to be revised independently of the predictive model.

Prediction and decision explanations. The framework provides two complementary forms of explanation. SHAP is used to identify the model features contributing to a vehicle's repair-proximity prediction, while the reasoning trace identifies the predictive and temporal evidence responsible for the recommended maintenance action. Keeping these explanation paths distinct allows users to distinguish why the model assessed a vehicle in a particular state from why the system recommended a particular level of maintenance attention.

Shared services and user interaction. Backend services associate each vehicle assessment with its prediction, temporal evidence, maintenance recommendation, and explanation outputs. The dashboard presents these records through fleet-level summaries and vehicle-level views, enabling users to identify priority vehicles and inspect the evidence underlying individual recommendations. A retrieval-augmented generation (RAG) assistant provides an additional interaction layer by retrieving relevant supporting documents and using the structured vehicle assessment as context for natural-language responses. The assistant does not independently generate telemetry values, predictions, or maintenance priorities, and does not override the deterministic recommendation produced by the reasoning engine.

This modular design supports evaluation at both component and system levels. Predictive performance can be assessed independently of reasoning behaviour, while the reasoning layer can be evaluated for the consistency and transparency of its recommendations. Integration evaluation can further examine whether the dashboard and RAG assistant present predictions, evidence, and recommendations consistently.

6.2 Predictive Model

The predictive layer is designed to estimate a vehicle's repair-proximity state from its operational history and static vehicle specifications. The current principal candidate is a vehicle-aware, multi-history XGBoost classifier.

XGBoost is being investigated as the main predictive approach because it is well suited to the project's engineered tabular representation, which combines current operational conditions, historical changes, and vehicle configuration. Random Forest provides a baseline for comparison, while alternative feature representations and training strategies will continue to be evaluated before final model selection.

For vehicle i at observation time t, the predictive input $\mathbf{x}_{i,t}$ contains only information available at or before that time. The candidate representation combines current operational values, previous observations, multi-readout changes, time-normalised rates of change, rolling summaries, and static vehicle specifications. The current prototype also uses vehicle-aware training preparation to reduce the disproportionate contribution of vehicles with long histories.

For each vehicle observation, the multiclass model produces five class scores:

$$\mathbf{s}_{i,t} = \left(s_{i,t,0}, s_{i,t,1}, s_{i,t,2}, s_{i,t,3}, s_{i,t,4}\right), \qquad \sum_{k=0}^{4} s_{i,t,k} = 1. \tag{2}$$

The predicted repair-proximity class is determined by the highest-scoring class:

$$\hat{y}_{i,t} = \underset{k\in\{0,1,2,3,4\}}{\arg\max} \; s_{i,t,k}. \tag{3}$$

Because C1–C4 represent mutually exclusive repair intervals within progressively shorter horizons, the class scores are also aggregated into four cumulative model risk scores:

$$\begin{aligned} \text{Risk48}_{i,t} &= \sum_{k=1}^{4} s_{i,t,k}, \\ \text{Risk24}_{i,t} &= \sum_{k=2}^{4} s_{i,t,k}, \\ \text{Risk12}_{i,t} &= s_{i,t,3} + s_{i,t,4}, \\ \text{Risk6}_{i,t} &= s_{i,t,4}. \end{aligned} \tag{4}$$

Risk48 therefore summarises model evidence associated with repair within the 48-unit horizon, while Risk6 corresponds to the most imminent C4 interval. These quantities are treated as model-derived risk scores rather than calibrated real-world probabilities of repair. Probability calibration will be evaluated as part of model development before the final interpretation of these scores is fixed.

A single current prediction may not fully represent how a vehicle's risk has developed. The proposed system therefore applies the same predictive pipeline to historical vehicle snapshots, using only the information available at each snapshot. This produces historical sequences of predicted states and risk scores.

These historical outputs are summarised into supporting indicators, including:

- **Fleet-relative risk:** the position of a vehicle's current risk score relative to other vehicles in the evaluated fleet;
- **Risk persistence:** whether elevated risk appears repeatedly across recent observations;
- **Relative deterioration:** whether model-derived risk is changing unusually quickly relative to other vehicles;
- **State trajectory and stability:** how recent predicted repair-proximity states have changed and how consistently they have been predicted; and
- **History sufficiency:** whether sufficient historical observations are available to support the assessment.

The current prototype examines up to five recent observations for persistence and deterioration analysis. Initial persistence and fleet-relative percentile thresholds are treated as prototype policy settings and will be evaluated during further development.

These indicators are derived from historical outputs of the same predictive model; they are not additional ground-truth labels or a separate temporal prediction model. Their purpose is to provide historical context for the knowledge-based reasoning layer.

6.3 Knowledge-Based Rule Engine

The proposed knowledge-based rule engine translates predictive and historical evidence into maintenance-priority recommendations. An explicit IF–THEN design is used because it provides transparent and auditable decision logic and allows maintenance policies to be reviewed or modified independently of predictive-model training.

Inputs to the reasoning layer include the current predicted repair-proximity state, fleet-relative risk, historical risk persistence, relative deterioration, recent predicted-state trajectory, and history sufficiency. The output consists of a recommended maintenance action, priority level, triggered rule identifier, and explanatory reasoning trace.

Table 4 summarises the current prototype policy. Rules are evaluated in order, and the first satisfied rule determines the recommendation.

Within the current prototype, high fleet-relative risk is defined using upper-percentile thresholds for the relevant horizon scores. Risk48, Risk12, and Risk6 use progressively stricter thresholds as the repair horizon becomes shorter. Persistent risk requires the corresponding high-risk condition to appear across a specified proportion of recent observations, while fast deterioration identifies unusually rapid changes in model-derived risk relative to the fleet.

These thresholds are prototype decision policies rather than established engineering thresholds or official Scania maintenance procedures. Their usefulness will therefore be evaluated through validation performance, intervention coverage, and the resulting maintenance workload.

Table 4: Prototype maintenance reasoning rules proposed for FleetCare AI. Rules are evaluated in order and thresholds remain subject to validation.

Rule	Condition	Recommended Action
TR0	C3/C4 with fewer than three historical readouts	Manual Review
TR1	C4 and Risk6 in the top 5%, supported by persistent Risk6 or fast Risk6/Risk12 deterioration	Priority Inspect
TR2	C3/C4 with Risk12 in the top 10%, persistent Risk12, fast Risk6/Risk12 deterioration, or a worsening class trajectory	Inspect
TR3	Below C3, but Risk6 is in the top 5%, persistent, and shows fast relative deterioration	Inspect
TR4	C2, or Risk12 in the top 10% with persistent Risk12 or fast Risk12 deterioration	Enhanced Monitor
TR5	C1, or Risk48 in the top 20% with persistent Risk48 or fast Risk48 deterioration	Monitor
TR6	No preceding rule is satisfied	Continue

Importantly, the rule engine does not alter the original C0–C4 prediction. A rule may change the recommended level of maintenance attention while retaining the model's original prediction. This separation allows predictive performance and maintenance policy to be evaluated independently and enables decision rules to be revised without retraining the predictive model.

6.4 Explainability

FleetCare AI proposes two complementary forms of explanation: model-level explanation and decision-level explanation.

Model-level explanation. SHAP will be used to examine the contribution of individual features to the predictive model's outputs. Global SHAP summaries will be used to identify which operational and historical features contribute most strongly to the model's predictions, while vehicle-level explanations will show which features contribute to an individual prediction.

Because the operational variables are anonymised, SHAP explanations will retain the published feature identifiers and describe their statistical role, such as a current level, recent change, or rolling variation. Feature contributions will not be interpreted as evidence of a specific mechanical fault or causal failure mechanism.

Decision-level explanation. The proposed knowledge-based reasoning layer will provide a separate reasoning trace for each maintenance recommendation. The trace will record the triggered rule and supporting evidence, such as the current predicted state, fleet-relative risk, persistence, deterioration, and the availability of historical evidence.

The two explanation mechanisms are intended to address different questions:

- SHAP: Why did the predictive model produce this prediction?
- Rule trace: Why did the system recommend this level of maintenance attention?

This distinction reflects the separation between predictive assessment and maintenance reasoning in the proposed system. A maintenance recommendation may incorporate temporal and fleet-relative evidence beyond the information represented by the current predicted class alone.

6.5 Application Layer

The proposed application layer will connect the modelling and reasoning pipeline to a fleet-management dashboard through structured backend services. A React-based frontend and Python FastAPI backend are currently proposed, subject to the team's final technology selection.

The dashboard is intended to provide both fleet-level and vehicle-level decision support. At fleet level, the primary view will summarise the distribution of recommended maintenance actions, allowing users to identify how many vehicles require different levels of attention.

At vehicle level, the interface is planned to present:

- the predicted C0–C4 repair-proximity state and class scores;
- Risk48, Risk24, Risk12, and Risk6 scores and fleet-relative rankings;
- recent predicted-state and risk trajectories;
- historical persistence and deterioration indicators;
- recommended maintenance action and priority;
- the triggered rule and supporting reasoning evidence; and
- SHAP-based feature explanations where available.

The backend will expose these outputs through structured vehicle records containing identifiers, prediction snapshots, risk indicators, historical evidence, recommendations, and explanation fields. This design keeps predictive and reasoning logic within the backend rather than requiring the frontend to reproduce model calculations or decision rules.

The initial prototype may serve precomputed batch outputs from the available dataset. This provides a practical route to an end-to-end demonstration while model and interface development continues. On-demand inference can be considered once the preprocessing and prediction interfaces have been stabilised.

6.6 RAG-Based Assistant

A retrieval-augmented generation (RAG) assistant is proposed as an additional conversational interface to FleetCare AI. Its purpose is to help fleet and maintenance managers retrieve and interpret information already available within the system rather than independently generate maintenance decisions.

The assistant is planned to combine two information sources:

- Structured FleetCare AI outputs: vehicle predictions, risk scores, historical indicators, recommendations, and rule traces supplied through the backend;
- Curated supporting knowledge: dataset documentation, project-authored policy descriptions, relevant technical literature, and reviewed explanatory materials.

For vehicle-specific questions, the assistant will retrieve the relevant structured vehicle record and, where useful, supporting documents. Responses should distinguish between numerical model outputs, policy-based recommendations, and general contextual knowledge.

Example questions include:

- “Which vehicles currently require the highest level of maintenance attention?”
- “Why was this vehicle assigned Priority Inspect?”
- “How has this vehicle’s predicted risk changed across recent observations?”
- “Which high-urgency vehicles have limited historical support?”

The language model will be used to retrieve, organise, and explain available information. It will not generate replacement telemetry, recalculate the predictive model, override the deterministic rule engine, or infer unsupported physical meanings for anonymised operational variables.

If the available evidence is insufficient to answer a question, the assistant will indicate that limitation rather than generate unsupported vehicle or maintenance information.

Together, the dashboard and RAG assistant provide complementary interfaces to the same underlying predictive and reasoning outputs. The dashboard supports structured fleet monitoring and prioritisation, while the assistant supports natural-language exploration of vehicle evidence and system knowledge.

7 Implementation

At the proposal stage, implementation has focused on establishing the technical feasibility of the core FleetCare AI workflow while defining a practical architecture for subsequent system integration. An initial Python-based modelling pipeline and prototype knowledge-based reasoning workflow have been developed, while the user-facing application and RAG components remain part of the next implementation stage.

7.1 Current Prototype

The current prototype covers the analytical workflow from historical vehicle data to maintenance recommendations. Operational readouts are processed chronologically, temporal features are constructed from available vehicle history, and multiclass models are used to estimate repair-proximity states. Model outputs are subsequently transformed into horizon-specific risk indicators and historical evidence for the prototype rule engine.

The reasoning prototype applies explicit IF–THEN rules to combine the current prediction with fleet-relative and historical evidence. It produces a recommended maintenance action together with the rule and evidence supporting that recommendation.

This implementation establishes the feasibility of the proposed hybrid architecture while keeping prediction and decision logic separate.

7.2 Application Architecture and Technology Stack

The proposed application follows a separated frontend–backend architecture. Predictive and reasoning logic remains within the Python service layer, while the frontend is responsible for presenting fleet priorities and vehicle-level evidence. This separation allows the modelling and reasoning components to evolve without embedding their logic directly in the user interface.

The proposed technology stack is summarised in Table 5.

Table 5: Proposed technology stack for the FleetCare AI prototype.

Layer	Technology	Role
Data and modelling	Python, Pandas, NumPy, Scikit-learn, XGBoost	Data preparation, feature engineering, model development and inference
Explainability	SHAP	Interpretation of model outputs at global and vehicle levels
Reasoning	Python rule engine	Applies explicit IF–THEN policies and generates reasoning traces
Backend	FastAPI	Provides structured fleet and vehicle information to user-facing services
Frontend	React	Provides fleet overview, filtering and vehicle-detail interfaces
RAG assistant	Retrieval pipeline and LLM	Supports grounded natural-language access to vehicle evidence and project knowledge

7.3 Frontend and Backend Design

The backend is planned to expose two primary forms of structured output: fleet-level summaries and individual vehicle records.

Fleet-level outputs will support maintenance prioritisation by summarising recommended actions and priority levels across the fleet. Vehicle-level outputs will provide more detailed information, including repair-proximity predictions, risk scores, historical evidence, recommended actions and reasoning traces.

The frontend is therefore designed around two corresponding views:

- Fleet overview: maintenance-action distribution, priority summary, search and filtering; and
- Vehicle detail: predicted repair-proximity state, risk indicators, historical trajectory, maintenance recommendation and supporting evidence.

The initial application will use precomputed batch predictions from the available dataset. This provides a practical route to an end-to-end prototype because the dataset is historical rather than a live telemetry stream. On-demand prediction can be considered after the preprocessing and service interfaces have been stabilised.

7.4 Implementation Challenges and Initial Resolutions

Several technical issues have already influenced the prototype design. Irregularly sampled vehicle histories are addressed through time-aware temporal features rather than assuming equal intervals between read-outs. Vehicle-aware data preparation is used to reduce the disproportionate influence of vehicles with long histories, while censoring-aware labels reduce the risk of treating unknown future outcomes as confirmed negative cases.

The implementation also separates machine-learning outputs from maintenance recommendations. This allows the predictive model and rule policy to be developed and evaluated independently and provides a clearer reasoning trace for the user.

Further implementation challenges, particularly application integration and RAG grounding, will be addressed during the next development stage.

8 Results and Progress

At the proposal stage, the project has established the main data, modelling, reasoning, and application requirements for FleetCare AI. Initial data analysis and prototype development have been used to assess the feasibility of the proposed workflow, while quantitative model evaluation and integrated application testing remain part of the next development stage. The results reported here therefore represent proposal-stage progress rather than final system performance.

8.1 Predictive Module Progress

The predictive task has been formulated as five-class repair-proximity classification, with vehicle history and static specifications used as predictive inputs. A censoring-aware target construction strategy, vehicle-level validation protocol, and time-aware feature representation have been defined based on the characteristics identified during data analysis.

A vehicle-aware, multi-history XGBoost classifier has been selected as the principal modelling candidate, with Random Forest retained as a baseline for subsequent comparison. The predictive output interface has also been defined to provide a C0–C4 prediction, class scores, and the derived Risk48, Risk24, Risk12, and Risk6 scores required by the downstream reasoning layer.

Figure 5 summarises the resulting predictive module specification. Model configuration and feature choices remain subject to validation, and the model-derived risk scores will not be interpreted as calibrated repair probabilities without supporting calibration evidence.

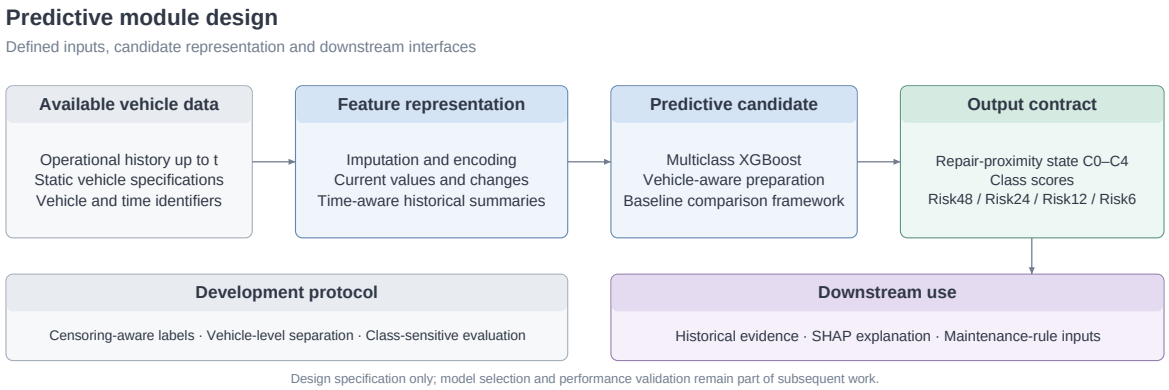


Fig. 5: Proposal-stage specification of the predictive module, showing the input categories, feature-processing direction, principal model candidate, and downstream output interface.

8.2 Reasoning and Explainability Progress

A prototype reasoning workflow has been defined to separate predictive assessment from maintenance recommendation. The workflow uses current model outputs together with temporal and fleet-relative evidence to support the maintenance-action categories defined in the proposed rule set.

The explanation requirements have also been specified at two levels. SHAP is proposed for examining feature contributions to predictive outputs, while explicit rule traces will record the evidence supporting maintenance recommendations. This establishes separate explanation paths for model behaviour and decision logic while recognising that anonymised features cannot support physical fault diagnosis.

8.3 Interface Design Progress

The user-interface requirements have been translated into three connected views: a fleet overview, a vehicle workspace, and an analytics and assistant workspace. Figure 6 shows the resulting conceptual interface. These wireframes define the information hierarchy and intended user journey; they do not represent implemented application screens or measured system outputs.

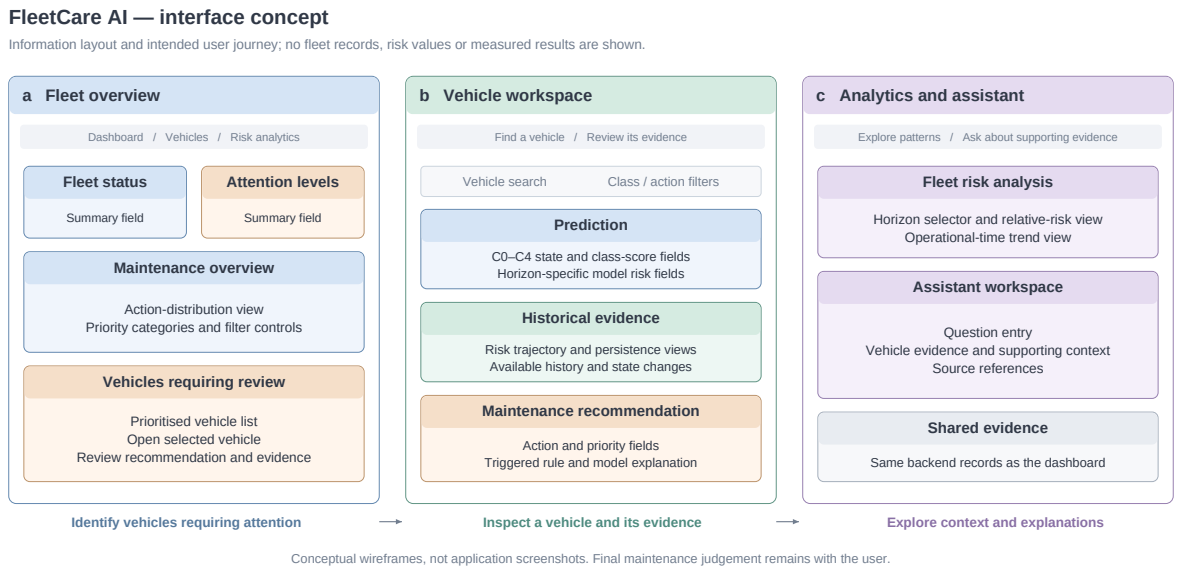


Fig. 6: Conceptual interface design for FleetCare AI: (a) fleet-level maintenance prioritisation, (b) vehicle-level assessment and supporting evidence, and (c) risk analysis and evidence-based assistance.

The fleet overview is intended to support identification and filtering of vehicles requiring different levels of maintenance attention. The vehicle workspace will bring together the current prediction, historical risk evidence, recommended action, and associated explanations. The analytics and assistant workspace will provide additional access to fleet-relative trends and grounded natural-language exploration of available vehicle and supporting information.

8.4 Integration Status and Subsequent Validation

A shared vehicle-assessment structure has been defined to connect the predictive model, reasoning layer, dashboard, and RAG assistant. Each assessment will associate an observation time with its predicted class, horizon-specific risk scores, temporal evidence, maintenance recommendation, and explanation references. User-facing components will consume the same structured backend outputs to reduce inconsistency between interfaces.

The next development stage will evaluate the predictive model, reasoning policies, and integrated application. Predictive evaluation will use vehicle-level separation and class-sensitive measures, including Macro-F1 and per-class precision and recall. Comparisons between current-value and temporal feature representations will examine the contribution of historical information. Rule evaluation will consider recommendation consistency, escalation workload, and behaviour when historical evidence is limited.

Application-level validation will examine whether vehicle assessments and supporting evidence are presented consistently across the backend, dashboard, and RAG assistant. Quantitative model results, rule evaluation outcomes, integrated application demonstrations, and representative vehicle cases will be reported in the final report.

9 Challenges and Roadblocks

The preliminary implementation has identified three main challenges that will guide the remaining development of FleetCare AI.

9.1 Predictive Performance and Data Characteristics

The primary modelling challenge is the combination of severe class imbalance, irregular vehicle histories and censored outcomes. C0 represents most available examples, while the near-repair classes contain substantially fewer vehicles. In addition, vehicles differ in both the number and spacing of their operational readouts.

These characteristics make minority-class prediction and reliable evaluation difficult. The remaining modelling work will therefore compare selected feature and training configurations using vehicle-level validation and class-sensitive metrics. Temporal representations will continue to account for actual operational-time differences rather than treating readouts as equally spaced.

9.2 Reasoning Validity and Explainability

The prototype rule engine provides transparent maintenance recommendations, but its percentile, persistence and deterioration thresholds are project-defined policies rather than validated maintenance standards. Their usefulness must therefore be evaluated against both near-repair coverage and the number of vehicles requiring attention.

Explainability is additionally constrained by the anonymised operational variables. SHAP can identify which model inputs contribute to a prediction but cannot determine their undocumented physical meaning or diagnose a specific mechanical fault.

The project will address these limitations by evaluating rule sensitivity and representative vehicle cases, while keeping model explanations separate from rule-based decision explanations. Recommendations will remain explicitly positioned as human decision support.

9.3 System Integration and Grounding

The remaining system-level challenge is to ensure that predictions, historical evidence, maintenance recommendations and explanations remain consistent across the backend, dashboard and RAG assistant.

A shared structured output schema will be used as the interface between components. Numerical vehicle information will be obtained from structured system outputs, while the RAG component will use retrieved documents for supporting contextual knowledge. This design reduces the risk of inconsistent values or unsupported generated vehicle information.

10 Future Work

The remaining project work will focus on completing and evaluating FleetCare AI as an integrated intelligent reasoning system. The main activities are organised into four areas.

10.1 Model and Reasoning Evaluation

Further modelling experiments will compare selected feature representations and training configurations under a consistent vehicle-level evaluation protocol. Evaluation will consider Macro-F1, minority-class performance, confusion patterns and multi-horizon risk-ranking measures.

The prototype maintenance rules will also be evaluated as a separate decision-support layer. Threshold sensitivity, near-repair coverage and the proportion of vehicles selected for different maintenance actions will be examined before the final prototype policy is selected.

10.2 Explainability and Vehicle Case Studies

Explainability work will extend to representative vehicle-level examples. Selected cases will combine the current prediction, historical risk trajectory, SHAP-based model explanation and rule-based reasoning trace.

Cases will include both successful and unsuccessful predictions so that the final evaluation reflects model limitations rather than presenting only favourable examples.

10.3 Frontend, Backend and RAG Development

The proposed React frontend and FastAPI backend will be developed into an integrated fleet-management application. Planned functionality includes a fleet-level maintenance-action overview, vehicle search and filtering, individual vehicle details, historical risk visualisations and reasoning evidence.

The RAG assistant will be connected to structured vehicle outputs and a curated knowledge collection. Its evaluation will examine retrieval relevance, source grounding, numerical consistency and handling of questions for which sufficient information is not available.

10.4 End-to-End Evaluation and Final Deliverable

The final stage will integrate the predictive model, knowledge-based reasoning layer, explainability components, dashboard and RAG assistant into a single demonstration workflow.

The final evaluation will report detailed predictive results, multi-horizon risk-ranking performance, maintenance-rule outcomes, representative explanation cases and user-facing system examples. Selected vehicles will be traced from operational history through prediction and reasoning to the final dashboard and conversational outputs.

The final deliverable will therefore evaluate not only whether the model can generate useful predictive evidence, but also whether that evidence can be presented as transparent and consistent decision support for fleet and maintenance managers.

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